JS CODE:

```
let books = JSON.parse(localStorage.getItem("books")) || [];
const form = document.getElementById("bookForm");
const bookList = document.getElementById("bookList");
// Save books in browser
function saveBooks() {
  localStorage.setItem("books", JSON.stringify(books));
}
// Add or Edit book
form.addEventListener("submit", function(event) {
  event.preventDefault();
  const editId = document.getElementById("editId").value;
  const title = document.getElementById("title").value.trim();
  const author = document.getElementById("author").value.trim();
  const isbn = document.getElementById("isbn").value.trim();
  const category = document.getElementById("category").value;
  const quantity = Number(document.getElementById("quantity").value);
  if (!title || !author || !isbn || !category ||
    !Number.isInteger(quantity) || quantity < 1) {
    alert("Please enter valid book details.");
    return;
  }
  if (editId) {
    const book = books.find(b => b.id === Number(editId));
    if (quantity < book.issued) {
      alert("Quantity cannot be less than issued copies.");
      return;
    }
```

```
        book.title = title;
        book.author = author;
        book.isbn = isbn;
        book.category = category;
        book.quantity = quantity;
        alert("Book updated successfully!");
    } else {
        if (books.some(b => b.isbn === isbn)) {
            alert("This ISBN already exists.");
            return;
        }
        books.push({
            id: Date.now(),
            title: title,
            author: author,
            isbn: isbn,
            category: category,
            quantity: quantity,
            issued: 0
        });
        alert("Book added successfully!");
    }
    saveBooks();
    form.reset();
    document.getElementById("editId").value = "";
    document.getElementById("submitBtn").textContent = "Add Book";
    displayBooks();
});
```

```
// Display books
function displayBooks() {
    const search = document.getElementById("search")
        .value.toLowerCase();
    const category = document.getElementById("filterCategory").value;
    const availability = document.getElementById("filterAvailability").value;
    bookList.innerHTML = "";
    const filteredBooks = books.filter(book => {
        const matchesSearch =
            book.title.toLowerCase().includes(search) ||
            book.author.toLowerCase().includes(search) ||
            book.isbn.toLowerCase().includes(search);
        const matchesCategory =
            category === "All" || book.category === category;
        const available = book.quantity - book.issued;
        const matchesAvailability =
            availability === "All" ||
            (availability === "Available" && available > 0) ||
            (availability === "Issued" && book.issued > 0);
        return matchesSearch && matchesCategory && matchesAvailability;
    });
    if (filteredBooks.length === 0) {
        bookList.innerHTML =
            `<tr><td colspan="7">No books found</td></tr>`;
    }
    filteredBooks.forEach(book => {
        const available = book.quantity - book.issued;
        bookList.innerHTML += `
```

```
      <tr>
        <td>${escapeHTML(book.title)}</td>
        <td>${escapeHTML(book.author)}</td>
        <td>${escapeHTML(book.isbn)}</td>
        <td>${escapeHTML(book.category)}</td>
        <td>${book.quantity}</td>
        <td class="status">${available}</td>
        <td>
          <button class="edit"
            onclick="editBook(${book.id})">Edit</button>
          <button class="delete"
            onclick="deleteBook(${book.id})">Delete</button>
          <button class="issue"
            onclick="issueBook(${book.id})">Issue</button>
          <button class="return"
            onclick="returnBook(${book.id})">Return</button>
        </td>
      </tr>
    `;
});
document.getElementById("totalTitles").textContent = books.length;
document.getElementById("totalCopies").textContent =
  books.reduce((sum, book) => sum + book.quantity, 0);
document.getElementById("availableCopies").textContent =
  books.reduce((sum, book) =>
    sum + book.quantity - book.issued, 0);
document.getElementById("issuedCopies").textContent =
  books.reduce((sum, book) => sum + book.issued, 0);
```

```
}
// Prevent HTML injection when displaying text
function escapeHTML(value) {
    return String(value).replace(/[&<>"']/g, character => ({
        "&": "&amp;",
        "<": "&lt;",
        ">": "&gt;",
        '"': "&quot;",
        "'": "&#39;"
    })[character]);
}
// Edit book
function editBook(id) {
    const book = books.find(b => b.id === id);
    document.getElementById("editId").value = book.id;
    document.getElementById("title").value = book.title;
    document.getElementById("author").value = book.author;
    document.getElementById("isbn").value = book.isbn;
    document.getElementById("category").value = book.category;
    document.getElementById("quantity").value = book.quantity;
    document.getElementById("submitBtn").textContent = "Update Book";
    window.scrollTo({ top: 0, behavior: "smooth" });
}
// Delete book
function deleteBook(id) {
    const book = books.find(b => b.id === id);
    if (book.issued > 0) {
        alert("Return all issued copies before deleting this book.");
```

```
        return;
    }
    if (confirm("Are you sure you want to delete this book?")) {
        books = books.filter(b => b.id !== id);
        saveBooks();
        displayBooks();
    }
}
// Issue book
function issueBook(id) {
    const book = books.find(b => b.id === id);
    if (book.quantity - book.issued > 0) {
        book.issued++;
        saveBooks();
        displayBooks();
        alert("Book issued successfully!");
    } else {
        alert("No copies available!");
    }
}
// Return book
function returnBook(id) {
   const book = books.find(b => b.id === id);
    if (book.issued > 0) {
        book.issued--;
        saveBooks();
        displayBooks();
        alert("Book returned successfully!");
```

```
    } else {
        alert("No copies have been issued!");
    }
}
// Search and filter
document.getElementById("search")
    .addEventListener("input", displayBooks);
document.getElementById("filterCategory")
    .addEventListener("change", displayBooks);
document.getElementById("filterAvailability")
    .addEventListener("change", displayBooks);
// Load saved books
displayBooks();
```